Imam Mohammad Ibn Saud Islamic University

College of Computer and Information Sciences

Computer Science Department

Course Title:	Design and Analysis of Algorithms
Course Code:	CS1211
Course Instructor:	Dr. Abdulrahman Albarrak
Project:	Maximize Cash Back Rewards
Semester:	Third Semester, 2024/2025
Due Date:	Sunday, May 18, 2025 (20-11-1446)
Marks:	20

Instructions:

1. Form a group of 2-5 students.
2. Read and understand the computational problem stated below.
3. Design an algorithm to solve the stated problem and write its pseudocode. **[4 marks]**
4. Analyse your algorithm running time, provide the running time function T(n) and an asymptotic upper bound using Big-O notation. **[4 marks]**
5. Implement your algorithm using Java or Python. **[4 marks]**
6. Test your algorithms and show the results **[3 marks]**
7. Submit your report (pseudocode, running time and asymptotic upper bound analysis) along with your .java or .py file via Blackboard before the project submission deadline.
8. Prepare a presentation that summarizes your work **[5 marks]**

Question	CLOs	Question Marks
Theoretical part	1.2	8
Practical part	2.2	7
Presentation	3.1	5
Total		**20**

Project Description

Many cards offer higher cash back rates in specific categories like dining, groceries, gas, or online purchases. This project focuses on designing and analysing algorithms to maximize the amount of cash back rewards while adhering to spending limits, taking into account categories that offer higher rewards and caps on rewards. There are multiple credit cards, each offering different cash back rates for different spending categories (e.g., Dining, Gas, Groceries), with some cards having spending caps for high-reward categories. The goal is to maximize the total cash back reward while adhering to the constraints (spending limits, categories, etc.).

Let's assume there are multiple credit cards that provide cashback for different spending categories. Each category has a specific cashback rate, and you want to optimize how you spend your money across these categories to maximize the total cashback.

Parameters:

1. **Spending Categories**: Each spending category has a cashback percentage (e.g., 2% for groceries, 3% for dining, etc.).
2. **Budget**: You have a fixed total amount of money you can spend (e.g., $5000 for the month).
3. **Cashback Rewards**: Different categories offer different cashback rates (e.g., 3% on groceries, 2% on dining, etc.).
4. **Spending Limits**: Some categories may have a spending limit or cap on how much you can spend to earn cashback (e.g., only the first $1000 spent in a category qualifies for cashback).

Create a system that allows users to input their credit cards and their cashback rewards program details. The system should calculate the optimal amount to spend on each card to maximize cashback rewards, considering spending limits and categories.

- Input: spending limit in different categories and multiple credit cards with different cashback rates and categories.
- Output: Optimal spending for each card and the total cashback earned.
- Constraints: Each card has different cashback categories (e.g., 5% on groceries, 2% on dining) and limits (e.g., max $500 on groceries).

Example Run

Spending:

Category	Amount Spent
Groceries	$600
Dining/Restaurants	$400
Gas	$200
Travel	$300
Online Shopping	$500
Other Purchases	$800

Student Name (in English): _____ **Student ID:** _____

Cash-back rates:

Card \ Category	Groceries	Dining	Gas	Travel	Online Shopping	other	Limit
Card 1	6%	2%	3%	1.5%	3%	1.5%	up to $1500
Card 2	5%	3%	2%	5%	2%	1.5$	up to $700
Card 3	2%	5%	4%	2%	5%	1%	up to $2500
Card 4	1.5%	2%	2%	4%	2$	2%	up to $3500

Summary of Rewards:

Category	Amount Spent	Card Used	Cash Back Earned
Groceries	$600	Card 1 Preferred (6%)	$36
Dining/Restaurants	$400	Card 3 (5%)	$20
Gas	$100	Card 3 (4%)	$4
Travel	$300	Card 2 (5%)	$12
Online Shopping	$500	Card 3 (5%)	$25
Other Purchases	$800	Card 4 (2%)	$16
Total Cash Back			**$113**

In this scenario, the optimal total cash back is **$113** based on the cards and categories that maximize rewards.

Student Name (in English): _____ **Student ID:** _____

Task 1: Theoretical part [ ] / 8 Marks

In the theoretical part, you are requested to:

a) Design two algorithms to maximize cash back rewards by dynamically selecting the best credit card for each spending category, given a set of cards and their corresponding reward percentages for different categories and limit in each card.
The stated problem should be solved using the following algorithms:

1. A brute-force algorithm.
2. A greedy algorithm or a dynamic-programming based algorithm.

In the report, present the designed algorithms and write their pseudocodes. **[4 Marks]**

b) Analyze the designed algorithms and compute an asymptotic upper bound using T(n) for the time complexity and the derived Big-O notation. Conduct a theoretical analysis of the time complexity of all designed algorithms. Compare and explain their time complexity. **[3 Marks]**

c) Compare the designed algorithms, discussing their strengths, weaknesses, and use cases. **[1 Mark]**

Task 2: Practical part [ ] / 7 Marks

In the practical part, you are requested to:

a) Implement the algorithms and present the code with clear comments. **[4 Marks]**

b) Test and analyze the implemented algorithms on different random datasets with varying sizes of spending limit and cards with different reward percentages. Show results in terms of execution time, space usage, and correctness of outputs. **[3 Marks]**

In the report, present the test results and analyze how the algorithms perform under different input sizes.

Student Name (in English): _____ **Student ID:** _____

Task 3: Presentation [ ] / 5 Marks

Summarize the entire project in a presentation, including

- Overview
- The problem definition,
- Solution designed approaches with pseudocodes,
- The time complexity analysis (execution time, space complexity)
- Results, and
- Conclusion with potential future work.

Deliverables

In the report:

1. **Algorithm Design:**
   - Explain the modelling of cards and spending limits (chosen data structures), designed algorithms, and how they solve the problem.
   - Provide mathematical formulations where necessary (e.g., greedy, dynamic programming).
   - Pseudocodes of the designed algorithm.
   - Time complexity analysis T(n) and O(n)
2. **Implementation:**
   - Implement the two algorithms
   - Include functions for adding new cards, spending limit, Categories, amount and reward percentages dynamically.
3. **Performance Analysis:**
   - Include performance benchmarks such as execution time, space complexity, and a comparison of the two designed algorithms in various scenarios.

In Final Presentation:

- Summarize the entire project in a presentation, including the problem definition, solution approach and results.

Collaboration: You must work in groups of *2-5 students*.

Submission: Submit through the Blackboard a compressed folder includes the report in addition to the source code files with the following naming schema: *CS1211_Project_Students' Names*.

Due time will be on Sunday, May 18, 2025 (20-11-1446)

Discussion time: Sunday and Monday (18-19 / May 2025)